A

Please insert in the circle the initial of your family name

Department of Physics
PCS 125 – Waves and Fields

Please Print Clearly:

Student: [redacted] [redacted]
Family Name *First Name*

Student Number: [redacted] **Student Section:** [redacted]

Mid -Term Test – February 28, 2014

Put a check mark in the box in front of the name of your professor:

Dr. Carvalho		Dr. Hu		Dr. Manzhos		Dr. Xu		Dr. Yuan	✓

Read the following attentively

- **Duration of the test: 1 hour and 30 minutes (90 minutes).**
- **The test consists of 1 full answer question (6 marks) and 12 multiple-choice questions (1 mark each). Answer ALL questions.**
- **For each multiple choice question:**
  - **You need to show, in the space provided in this booklet, a justification for your answer. If no work or correct argument is shown, the resultant mark may be zero, even if the answer in the bubble sheet is correct.**
  - **The work shown in your exam booklet may be checked, <u>at least for one</u> of the questions, selected randomly.**
  - **<u>Transfer your choices to the bubble sheet only when you're sure of them.</u>**
  - **Some answers may be rounded off.**
- **For the full answer question, show, in the space provided, all your work. For part marks to be awarded, the steps leading to the answers in each part of the problem have to be organized, unambiguous and clear.**
- **Pagers and cell phones must be silenced and placed out of reach. Hats and earphones are not allowed. If a phone rings, its owner is responsible for disrupting the exam and the phone will be confiscated or the owner will be asked to leave the room resulting in a zero on the test.**
- **Sharing of pens, erasers and calculators is not permitted.**
- **Talking to another student or glancing over another student's paper is not permitted and may result in a charge of academic misconduct.**
- **Formula sheet provided with the test booklet. No other sheets allowed.**
- **The only calculators allowed are either the Sharp EL-546 or Casio FX-991. If you are found to have another type of calculator (even if not programmable) <u>it will be confiscated</u> for the duration of the test.**

Student signature (no signature, no mark!)

$k = 2.5$ N/m

$V = 0.2$ m/s at $x = -0.05$ m

$m = 0.1$ kg

$2.5 + 9 = 11.5$

Full Answer Question (6 marks)

A 100g block attached to a spring of spring constant 2.5 N/m oscillates horizontally on a frictionless table. Its velocity is v=20 cm/s when $x = -5.0$ cm. Answer the following questions and insert your final answer in the boxes provided. DO NOT FORGET UNITS.

$\omega = 2\pi f$

$f = \frac{\omega}{2\pi}$

$\omega = \sqrt{\frac{k}{m}}$

$f = \frac{5}{2\pi}$

a) (2 marks) What is the amplitude of the SHM oscillations?

$k_{\text{non}} = 25$

$t = 4$

$y = A\sin(25x - 5t)$

$V = -5A\cos(25x - 5t)$

$0.2 = -5A\cos(25(-0.05) - 5(4))$

$0.2 = -5A\cos(-2.25)$

$0.2 = 3.688A$

$A = 0.054$ m

$\boxed{\omega = 5}$ +0.5

$V = \lambda f$

$\boxed{\lambda = 0.251327\,m}$

$0.2 = \lambda \frac{5}{2\pi}$

$A = 0.054$ m

b) (2 marks) What is the block's maximum acceleration?

$a_{max} = A\omega^2$ +2.0

$= 0.054(5)^2$

$\boxed{a_{max} = 1.35 \text{ m/s}^2}$

$a = 1.35$ m/s^2

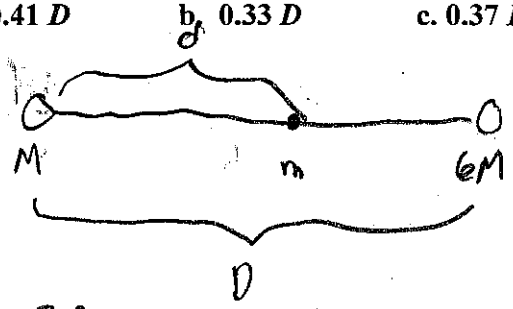
c) (2 marks) What is the speed of the block when it is at x=3 cm?

$v =$

MULTIPLE CHOICE QUESTIONS
(each question is worth 1 mark) Show your work in the space provided.

1. Two stars of masses $m_1=M$ and $m_2=6M$ are separated by a distance D. Determine the distance (measured from M) to a point at which the net gravitational force on a third mass, m_3 would be zero.

a. 0.41 *D* **b. 0.33 *D*** **c. 0.37 *D*** **d. 0.29 *D*** **e. 0.14 *D***



$D^2 - 2Dd - 7d^2 = 0$

$F_{net_3} = F_{g_1} + F_{g_2} = 0$

$\frac{GMm}{d^2} - \frac{G6Mm}{(D-d)^2} = 0$

$\frac{\cancel{GMm}}{d^2} = \frac{6\cancel{GMm}}{(D-d)^2}$

$(D-d)^2 = 6d^2$

$D - d = \sqrt{6}\,d$

$D = \sqrt{6}\,d + d$

$D = 3.45\,d$

$\boxed{d = 0.29D}$

2. An object attached to the bottom of a vertical spring with spring constant 18.0 N/m is set into simple harmonic motion with amplitude 0.400 m. It is observed that the maximum speed of the object is 90.0 cm/s. What is the weight of the object?

a. 34.8 N **b. 31.4 N** **c. 0.314 N** **d. 78.4 N** **e. 87.1 N**

$K = 18$ N/m

$A = 0.400$ m

$V_{max} = 0.9$ m/s

$V_{max} = A\omega$

$\omega = \frac{V}{A}$

$\omega = 2.25$ rad/s

$\omega = \sqrt{\frac{k}{m}}$

$2.25 = \sqrt{\frac{18}{m}}$

$m = \frac{18}{5.0625}$

$m = 3.555$

$F = mg$

$\boxed{34.8\ N}$

3. A simple pendulum makes 150 complete oscillations in 5.0 min at a location where $g = 9.8\ m/s^2$. What is the length of this pendulum?

a. 3.1 m **b. 0.50 m** **c. 9.7 m** **d. 0.99 m** **e. 0.13 m**

$\frac{150}{(5 \cdot 60)} = \frac{150\ rev}{300\ sec} = f = \boxed{0.5\ Hz}$

$\omega = 2\pi f$

$\boxed{\omega = \pi}$

$\frac{1\ sec}{rev} = T$ $\frac{1\ rev}{sec} = f$

$\omega = \sqrt{\frac{g}{L}}$

$\pi^2 = \frac{g}{L}$

$L = \frac{g}{\pi^2}$

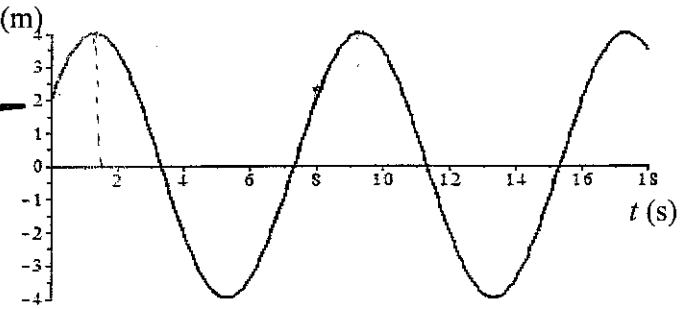
$\boxed{L = 0.99\ m}$

4. The simple harmonic motion of an object is described by the graph shown in the figure. What is the phase constant for this motion if the position of the object varies with time according to $x = A\cos(\omega t + \phi_0)$?

$T = 8\ s$

$\Delta t = 1.5\ s$

2 m



a. $-\pi/3$ **b. $2\pi/3$** **c. $-2\pi/3$** **d. $\pi/3$** **e. none of these (give the value)**

$\phi = \frac{\Delta t}{T} 2\pi$

$= \left(\frac{3}{2} \div 8\right) 2\pi$

$= \frac{3 \cdot 2\pi}{16}$

$= \frac{6\pi}{16}$

$\phi = \boxed{\frac{3\pi}{8}}$

$\phi = \frac{3\pi}{8}$

5. The tension in a taut rope is increased by a factor of 9 and its linear density is decreased by a factor of 4. How does the speed of the wave pulses on the rope change?

a. The speed remains the same

b. The speed is increased by a factor of 3/2

c. The speed is reduced by a factor of 2/3

d. The speed is increased by a factor of 6

e. The speed does not change

$$V = \sqrt{\frac{T}{\mu}}$$

$$V_2 = \sqrt{9T \div 0.25\mu}$$

$$= \sqrt{\frac{36T}{\mu}}$$

$$= 6\sqrt{\frac{T}{\mu}}$$

$$\boxed{V_2 = 6V_1}$$

6. The equation of a wave, traveling along the positive x direction with an amplitude of 2 cm, a frequency of 440 Hz, and a speed 330 m/s is given in SI units by

~~a. $y = 0.02\cos[880\pi(x/330 + t)]$~~

~~b. $y = 2\cos[880\pi(x/330 - 440t)]$~~

c. $y = 0.02\cos[880\pi(x/330 - t)]$

~~d. $y = 0.02\cos[2\pi(x/330 - 440t)]$~~

~~e. $y = 0.02\cos(2\pi(x/330 + 400t)]$~~

$\boxed{A = 0.02\ m}$

$f = 440\ Hz$

$\boxed{\omega = 880\pi\ Hz}$

$V = 330\ m/s$

$k = \frac{2\pi}{\lambda}$

$V = \lambda f$

$k = 2\pi$

$\lambda = \frac{330}{440}$

$y = 0.02\cos\left(\frac{880\pi x}{330} - 880\pi\right)$

$2\pi \div \frac{330}{440} = k = \boxed{\frac{880\pi}{330}}$

7. As a certain sound wave travels through some gas of density 1.40 kg/m^3, the displacement wave function is given by $s(x, t) = 2.54\times10^{-6} \sin(\pi x/2 + 200\pi t)$ in SI units. What is the pressure amplitude?

$\omega = -200\pi$

$k = \frac{\pi}{2}$

a. 0.140 N/m^2 **b. 0.278 N/m^2** **c. 0.651 N/m^2** **d. 0.763 N/m^2** **e. 0.894 N/m^2**

$V = \lambda f$

$\Delta P_{max} = \rho V \omega S_{max}$

$= (1.4)(400)(200\pi)(2.54\times10^{-6})$

$= 0.894$ N/m^2

$\boxed{f = 100 \text{ Hz}}$

$\omega = 2\pi f$

$\frac{200\pi}{2\pi}$

$\frac{\pi}{2} = \frac{2\pi}{\lambda}$

$\boxed{\lambda = 4}$

8. Sitting in a bus moving at 25.0 m/s, you hear a frequency of 580 Hz from the siren of an approaching ambulance. After the ambulance passes, the observed frequency of the siren is 410 Hz. Determine the ambulance's speed from these observations. (Take the speed of sound to be 343 m/s.)

a. 82.8 m/s **b. 58.9 m/s** **c. 70.5 m/s** **d. 34.3 m/s** **e. 71.4 m/s**

$580 = \left(\frac{343 - 25}{343 - V_s}\right) f$

$f = 580\left(\frac{343 - V_s}{318}\right)$

$410 = \left(\frac{343 + 25}{343 + V_s}\right) f$

$f = 410\left(\frac{343 + V_s}{368}\right)$

$410\left(\frac{343 + V_s}{368}\right) = 580\left(\frac{343 - V_s}{318}\right)$

$1.114(343 + V_s) = 1.824(343 - V_s)$

$382.1 + 1.114V_s = 625.632 - 1.824V_s$

$2.938V_s = 245.532$

$\boxed{V_s = 82.8 \text{ m/s}}$

Discard wrong scenario

9. The path difference between two waves emitted by coherent sources is 5m. If the wavelength of the waves is 4m, what is the phase difference ~~(in degrees)~~?

a. 90 b. 400 c. 1.57 d. 7.85 e. 15

$\Delta r = 5m$

$\lambda = 4m$

$2\pi = 360$

$\phi = \frac{\Delta r}{\lambda} = 1.25$

$0.25 \cdot 360 = 90°$

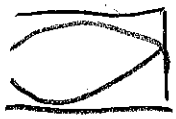
10. A tube open at one end and closed at the other end produces sound having a fundamental frequency of 350 Hz. If you now open the closed end, the fundamental frequency becomes

a. 87.5 Hz b. 175 Hz c. 350 Hz d. 700 Hz e. 1400 Hz

$n = 1$

$\frac{nV}{4L} = f$ $\frac{nV}{2L} = f_2$

opn cl



$\frac{V}{L} = 4f_1$ $\left(\frac{1400}{2}\right) = f_2$

$\frac{V}{L} = 1400$ $f_2 = 700$ Hz

$\frac{0.015\ g}{cm} \cdot \frac{1\ kg}{1000\ g} \cdot \frac{100\ cm}{1\ m} = 0.0015\ kg/m$

11. A string is stretched and fixed at both ends, 200 cm apart. If the density of the string is 0.015 g/cm, and its tension is 600 N, what is the wavelength (in cm) of the first harmonic?

a. 600 **b. 400** **c. 800** **d. 1000** **e. 200**

$\mu = 0.015\ g/cm = 0.0015\ kg/m$

$T = 600\ N$

$n = 1$

$L = 2m$

$V = \sqrt{\frac{T}{\mu}} = \boxed{632.46\ m/s}$

$V = \lambda f$

$\lambda = 800\ cm$

? $\boxed{f = \frac{nV}{4L} \quad 79\ Hz}$

12. Assume that a change in the sound source reduces the wavelength of a sound wave in air by a factor of 2, what happens to its speed?

a. It increases by a factor of 4.

b. It increases by a factor of 2.

c. It is unchanged.

d. It decreases by a factor of 2.

e. It changes by an unpredictable factor.

$V_1 = \lambda f$

$V_2 = \frac{\lambda f}{2}$

$\boxed{V_2 = \frac{V_1}{2}}$

↳ Assum freq won't changed

↳ $\lambda_2 = \frac{\lambda_1}{2}$